EE 66 Signals, Dynamics, and Information
Fall 2026 Homework 2

This homework is due Friday, September 11, 2026, at 23:59.

Submission Format

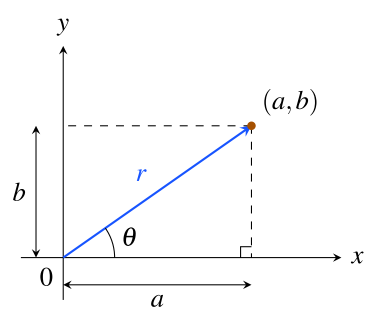
Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned).

On occasion, a problem set contains one more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

Submit the file to the appropriate assignment on Gradescope.

1. Complex Subspaces

Recall (from your physics classes, or vector calculus), we can describe a 2D vector as a point in 2D space. Sometimes we describe that point in "cartesian" coordinates, breaking it down into components along the orthogonal x and y axes. But other times, it is more convenient to describe the same vector in "polar" coordinates; here, we use its length r, and the angle θ formed between it and the positive x axis. Convince yourself that both formats can represent any point in 2D space equally well!



We use a *very similar* "change of coordinates" to represent complex numbers in two different ways:

$$z = a + bi \iff z = r\mathrm{e}^{i\theta}$$

In this problem, we associate with each complex number $z = a + bi$ a real vector $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$. In each part, you will consider a particular set of complex numbers. Determine if the associated set of vectors in $\mathbb{R}^2$ is a subspace. Justify your conclusion; and if it *is* a subspace, provide a basis for it.

Hint: try drawing each set of points in the complex plane, and justify your conclusion graphically.

(a) $S = \left\{v \mid v = r\mathrm{e}^{i\pi/3},\ r \in \mathbb{R}\right\}$.

(b) $T = \left\{v \mid v = 3\,\mathrm{e}^{i\theta},\ \theta \in [0, 2\pi)\right\}$.

(c) $U = \{v \mid v = bi,\ b \in \mathbb{R}\}$.

(d) $V = \{v \mid v = a + 2i,\ a \in \mathbb{R}\}$.

2. Operations on Subspaces

Let $\mathbb{V}$ be a vector space with subspaces $\mathbb{S}$ and $\mathbb{T}$.

Hint: in this problem, you are asked to write some more complicated proofs than you've encountered so far in this class. Fear not! It is generally good proof technique to start by simply writing down all the definitions you know. For example, what can you say about two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{S}$ using the fact that $\mathbb{S}$ is a subspace? Then think about how can you use these things you know to be true to show the desired conclusion.

(a) Consider the set of vectors $\mathbb{S}+\mathbb{T} := \{\mathbf{s}+\mathbf{t} \mid \mathbf{s} \in \mathbb{S}, \mathbf{t} \in \mathbb{T}\}$. Show that $\mathbb{S}+\mathbb{T}$ is a subspace of $\mathbb{V}$.

(b) Consider the intersection of $\mathbb{S}$ and $\mathbb{T}$: $\mathbb{S} \cap \mathbb{T} = \{\mathbf{v} \in \mathbb{V} \mid \mathbf{v} \in \mathbb{S}, \mathbf{v} \in \mathbb{T}\}$. Show that this is a subspace. *Hint: if $v \in \mathbb{S} \cap \mathbb{T}$, this means $v \in \mathbb{S}$ AND $v \in \mathbb{T}$*

(c) As before, $\mathbb{S}$ and $\mathbb{T}$ are subspaces of the vector space $\mathbb{V}$. Now, assume $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$. Let $B_S = \{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_k\}$ be a basis for $\mathbb{S}$, and $B_T = \{\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_n\}$ be a basis for $\mathbb{T}$.

  i. Show that $B_S \cup B_T = \{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_k, \mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_n\}$ is a basis for $\mathbb{S}+\mathbb{T}$.
  Hint 1: To show that these vectors span $\mathbb{S}+\mathbb{T}$, think about how you can write $\mathbf{s} \in \mathbb{S}$ in terms of the basis vectors in B_S? A similar thing can be done for any $\mathbf{t} \in \mathbb{T}$.
  Hint 2: To show linear independence, note that since $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$, if $\mathbf{s} \neq \mathbf{0}$ and $\mathbf{t} \neq \mathbf{0}$, then $\mathbf{s}+\mathbf{t} \neq \mathbf{0}$.

  ii. What is $\dim(\mathbb{S}+\mathbb{T})$ in terms of $\dim(\mathbb{S})$ and $\dim(\mathbb{T})$?

(d) (**Optional**) For arbitrary subspaces $\mathbb{S}$ and $\mathbb{T}$ of vector space $\mathbb{V}$, show that $\dim(\mathbb{S}+\mathbb{T}) = \dim(\mathbb{S}) + \dim(\mathbb{T}) - \dim(\mathbb{S} \cap \mathbb{T})$. Observe that If $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$, then $\dim(\mathbb{S} \cap \mathbb{T}) = 0$ and we recover the result from part (c).

3. Mechanical Basis

(a) Consider the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

i. Are $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent? Explain why.

ii. Can $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ form a basis for the vector space $\mathbb{R}^4$? Provide a succinct, yet clear and convincing explanation.

(b) Consider the set of vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

i. Explain why the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ is or is not a basis for $\mathbb{R}^3$.

ii. Consider $\mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}$.

Express $\mathbf{x}$ as a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$. Is this linear combination unique? If so, explain. If not, give an alternative linear combination.

4. Projections

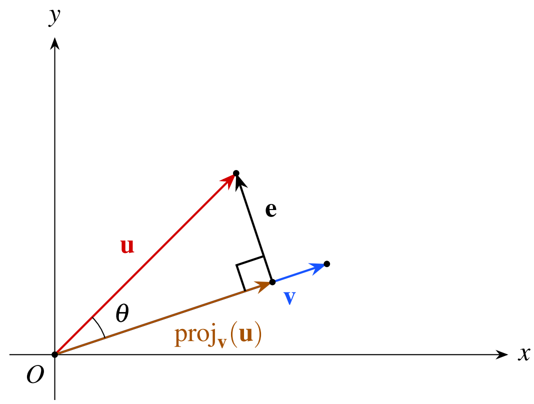


Figure 1: $\text{proj}_{\mathbf{v}}(\mathbf{u})$ visualized in $\mathbb{R}^2$

Figure 1 shows an example of an important concept in linear algebra known as **vector projection**, the act of "projecting" a vector onto another vector. This allows one to deconstruct any vector $\mathbf{u}$, given another vector $\mathbf{v}$, into two components:

- $\text{proj}_{\mathbf{v}}(\mathbf{u})$: a vector representing the component of $\mathbf{u}$ that points along another vector $\mathbf{v}$. This is also known as the **projection** of $\mathbf{u}$ onto $\mathbf{v}$.
- $\mathbf{e}$ is equal to $\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})$. In vector projection, $\mathbf{e}$ will be constructed such that $\|\mathbf{e}\|$ is minimized. Because of this, $\mathbf{e}$ will always be orthogonal to both $\mathbf{v}$ and $\text{proj}_{\mathbf{v}}(\mathbf{u})$. Intuitively, can you understand why? We will formally prove this later in the class.

In this problem, we will geometrically derive the formula for $\text{proj}_{\mathbf{v}}(\mathbf{u})$, and motivate the intuition behind vector projection.

(a) Using Figure 1, calculate $\|\text{proj}_{\mathbf{v}}(\mathbf{u})\|$. You may use θ, $\|\mathbf{u}\|$, and $\|\mathbf{v}\|$ in your answer.

(b) Now, using your answer to part (a), express $\text{proj}_{\mathbf{v}}(\mathbf{u})$ in terms of $\theta, \|\mathbf{u}\|, \|\mathbf{v}\|, \mathbf{u}$, and $\mathbf{v}$.

(c) Finally, express $\text{proj}_{\mathbf{v}}(\mathbf{u})$ in terms of $\mathbf{u}, \mathbf{v}$, and relevant inner products. You may not use norms or θ in your expression.

Hint: How can we rewrite terms that include norms and θ in terms of inner products?

5. Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality states that for two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$|\langle \mathbf{v}, \mathbf{w} \rangle| = |\mathbf{v}^\top \mathbf{w}| \leq \|\mathbf{v}\| \cdot \|\mathbf{w}\|$$

In this problem we will prove the Cauchy-Schwarz inequality for vectors in $\mathbb{R}^2$.

Take two vectors: $\mathbf{v} = r \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ and $\mathbf{w} = t \begin{bmatrix} \cos\phi \\ \sin\phi \end{bmatrix}$, where $r > 0$, $t > 0$, $\theta \in [0, 2\pi]$, and $\phi \in [0, 2\pi]$ are scalars. Make sure you understand why any vector in $\mathbb{R}^2$ can be expressed this way and why it is acceptable to restrict $r, t > 0$.

(a) In terms of some or all of the variables r, t, θ, and ϕ, what are $\|\mathbf{v}\|$ and $\|\mathbf{w}\|$?
Hint: Recall the trigonometric identity: $\cos^2 x + \sin^2 x = 1$.

(b) In terms of some or all of the variables r, t, θ, and ϕ, what is $\langle \mathbf{v}, \mathbf{w} \rangle$?
Hint: The trigonometric identity $\cos a \, \cos b + \sin a \, \sin b = \cos(a - b)$ *may be useful.*

(c) Now we will show that the Cauchy-Schwarz inequality holds for any two vectors in $\mathbb{R}^2$. To do this we must show that $\langle \mathbf{v}, \mathbf{w} \rangle$ is upper-bounded by $\|\mathbf{v}\|\|\mathbf{w}\|$ and lower-bounded by $-\|\mathbf{v}\|\|\mathbf{w}\|$. For this part, show that $\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\|\|\mathbf{w}\|$.
Hint: consider your results from part (b). Also recall $\cos x \leq 1$

(d) Now show that $\langle \mathbf{v}, \mathbf{w} \rangle \geq -\|\mathbf{v}\|\|\mathbf{w}\|$.
Hint: consider your results from part (b). Also recall $\cos x \geq -1$.

(e) Note that the inequality states that the absolute value of the inner product of two vectors must be less than *or equal to* the product of their magnitudes. What conditions must the vectors satisfy for the equality to hold? In other words, when is $|\langle \mathbf{v}, \mathbf{w} \rangle| = \|\mathbf{v}\| \cdot \|\mathbf{w}\|$?

6. Triangular Reasoning

(a) Let the vectors $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ serve as the legs of a triangle. What is the length of the hypotenuse of this triangle?

(b) Let us define the following two vectors:

$$\mathbf{x} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

What is the distance between these two vectors?
Hint: It may be useful to recognize that $||\mathbf{x}|| = 3\sqrt{2}$ *and* $||\mathbf{y}|| = 4\sqrt{2}$. *What is the angle between the two vectors?*

(c) Use the Cauchy-Schwarz inequality to verify (i.e. prove or derive) the triangle inequality:

$$||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$$

7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

(a) Did you work on this homework alone or in a group? Did this work well for you? *We would encourage you next week to try working with others if you usually work alone, and vice versa.*

(b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Please also include AI tools, if you've used any.

(c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

(d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

(e) Is there anything else you would like to share?

